ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems

Model answers to tutorial questions

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Topic 2: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

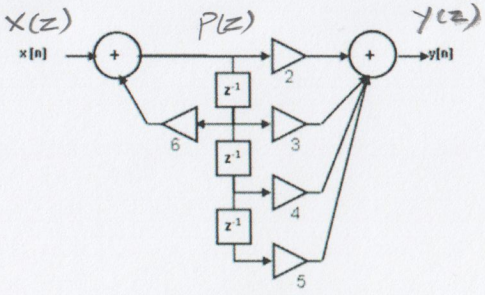
1. What is the equivalent transfer function of the block diagram below? (Tick the correct answer)

(a) $H(z) = \frac{z(z^{-1}+3)(z^{-1}+4)(z^{-1}+5)}{6-z^{-1}}$

✓(b) $H(z) = \frac{2+3z^{-1}+4z^{-2}+5z^{-3}}{1-6z^{-1}}$

(c) $H(z) = \frac{1+6z^{-1}}{2+3z^{-1}+4z^{-2}+5z^{-3}}$

(d) $H(z) = \frac{2+3z^{-1}+4z^{-2}+5z^{-3}}{1+6z^{-1}}$



$$\frac{P(z)}{X(z)} = \frac{1}{1-6z^{-1}}$$

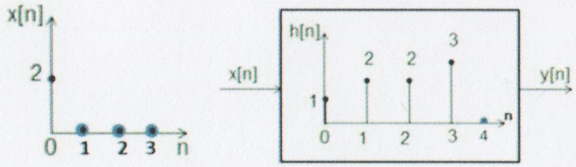
$$\frac{Y(z)}{P(z)} = 2+3z^{-1}+4z^{-2}+5z^{-3}$$

$$\therefore \frac{Y(z)}{X(z)} = \frac{Y(z)}{P(z)} \cdot \frac{P(z)}{X(z)}$$

$$= 2+3z^{-1}+4z^{-2}+5z^{-3} \times \frac{1}{1-6z^{-1}}$$

2. The impulse response ,h[n], of a discrete time system is given below. The output signal, y[n] , for the input signal, x[n] is given by: (Tick the correct answer)

(a) {2 0 0 0 0}
(b) {1 2 2 0 0}
(c) {3 4 5 2 0}
✓(d) {2 4 4 6 0}
(e) None of the above



$$y(n) = x(n) * h(n)$$

$$= 2\delta(n) * h[n]$$

↑ impulse (unit amplitude)

$$= 2\,h[n]$$

$$= \{2\ 4\ 4\ 6\ 0\}$$

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3. The impulse response of the system with a transfer function $H(z) = \frac{\alpha+\alpha z^{-1}}{1-\alpha z^{-1}}$ is given by: (Tick the correct answer)

   (a) $\alpha^{n-1} + \alpha \quad n < 0$

   ✓(c) $\alpha^{n+1}\, u[n] + \alpha^{n}\, u[n-1]$

   (b) $\alpha^{n}\, u[n] + \alpha^{n-1}\, u[n-1]$

   (d) None of the above

$H(z) = \frac{\alpha}{1-\alpha z^{-1}} + \frac{\alpha \cdot z^{-1}}{1-\alpha z^{-1}}$ — use inverse z-transform

inverse z-transform →

$h[n] = \alpha \cdot (\alpha)^{n} u[n] + \alpha \cdot (\alpha)^{n-1} u[n-1]$

$h[n] = \alpha^{n+1} u[n] + \alpha^{n} u[n-1]$

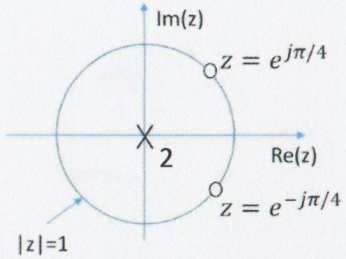
4. The pole-zero pattern of a system has zeros on the unit circle as show below. The system function is given by: (Tick the correct answer)

   (a) $H(z) = 1 + \frac{1}{\sqrt{2}} z^{-1} + z^{-2}$

   (b) $H(z) = z - \frac{1}{\sqrt{2}} + z^{-1}$

   (c) $H(z) = z^2 - 1 + z^{-2}$

   ✓(d) $H(z) = 1 - \sqrt{2}\, z^{-1} + z^{-2}$



$H(z) = \frac{(z - 1 \cdot e^{j\frac{\pi}{4}})(z - 1 \cdot e^{-j\frac{\pi}{4}})}{z^2}$

$= z^2 - \underbrace{(e^{j\frac{\pi}{4}} + e^{-j\frac{\pi}{4}})}_{=2\cos\frac{\pi}{4}} z + 1 \Big/ z^2$

$H(z) = \frac{z^2 - 2\cos\frac{\pi}{4} z + 1}{z^2} = \frac{z^2 - \sqrt{2} z + 1}{z^2}$

$H(z) = 1 - \sqrt{2} z^{-1} + z^{-2}$

5. The DFT of a real signal is given below. Therefore, the signal energy is equal to: (Tick the correct answer)

   (a) 0

   (b) 5

   (c) 7

   ✓(d) None of the above

DFT: [1, $\sqrt{2}$, -1, B, -7, -j2, C, $\sqrt{2}$]

k=0 ↑

use conjugate symmetry of DFT to find B & C

[1 $\sqrt{2}$ -1 B -7 -j2 C $\sqrt{2}$]

∴ B = j2

C = -1

$X[k] = [\underset{k=0}{1} \ \sqrt{2} \ -1 \ j2 \ -7 \ -j2 \ -1 \ \sqrt{2}]$ (k = 7)

signal energy $= \frac{1}{8} \sum |X[k]|^2$

$= \frac{1}{8} [1^2 + \sqrt{2}^2 + (-1)^2 + 2^2 + 7^2 + 2^2 + 1^2 + (\sqrt{2})^2]$

$= 8$

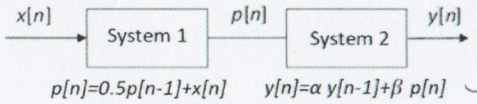
Topic 2: Tutorial Questions (Level 2)

SOLO – Level 2

1. Consider the cascade of the following two systems as depicted below. The difference equation relating $x[n]$ of $y[n]$ is given by:

$$y[n] = x[n] + \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2]$$

Determine α and β.



$p[n] = 0.5\,p[n-1] + x[n] \;\therefore\; \frac{P(z)}{X(z)} = \frac{1}{1-0.5z^{-1}}$

$y[n] = \alpha\, y[n-1] + \beta\, p[n] \rightarrow \frac{Y(z)}{P(z)} = \frac{\beta}{1-\alpha z^{-1}}$

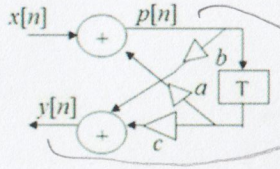
$\therefore \frac{Y(z)}{X(z)} = \frac{Y(z)}{P(z)} \cdot \frac{P(z)}{X(z)} = \frac{\beta}{1-\alpha z^{-1}} \cdot \frac{1}{1-0.5z^{-1}}$

$\therefore \boxed{\frac{Y(z)}{X(z)} = \frac{\beta}{1-(\alpha+0.5)z^{-1}+0.5\alpha z^{-2}}} \quad -(1)$

$\frac{Y(z)}{X(z)} = \frac{1}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}} \quad -(2)$

Equate (1) & (2) $\boxed{\beta = 1}$; $0.5\alpha = \frac{1}{8} \;\therefore\; \boxed{\alpha = \frac{1}{4}}$

2. For the block diagram realisations given below, develop the relation between y[n] and x[n] in each case.



(a)

$P(z) = X(z) + a z^{-1} P(z) \Rightarrow \frac{P(z)}{X(z)} = \frac{1}{1-az^{-1}}$

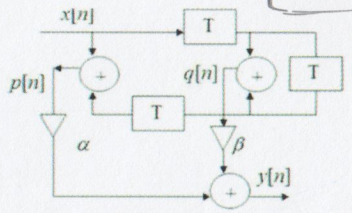
$Y(z) = b\,P(z) + c\,z^{-1} P(z) \Rightarrow \frac{Y(z)}{P(z)} = b + cz^{-1}$

$\Rightarrow \frac{Y(z)}{X(z)} = \frac{b+cz^{-1}}{1-az^{-1}}$

$Y(z)[1-az^{-1}] = X[z](b+cz^{-1})$

$Y(z) = bX(z) + cz^{-1}X(z) + az^{-1}Y(z)$

$\boxed{y[n] = b\,x[n] + c\,x[n-1] + a\,y[n-1]}$



(b)

$q[n] = x[n-1] + x[n-2]$

$p[n] = x[n] + x[n-3]$

$y[n] = \alpha\, p[n] + \beta\, q[n] = \alpha\{x[n] + x[n-3]\} + \beta\{x[n-1] + x[n-2]\}$

$\boxed{\therefore y[n] = \alpha\, x[n] + \beta\, x[n-1] + \beta\, x[n-2] + \alpha\, x[n-3]}$

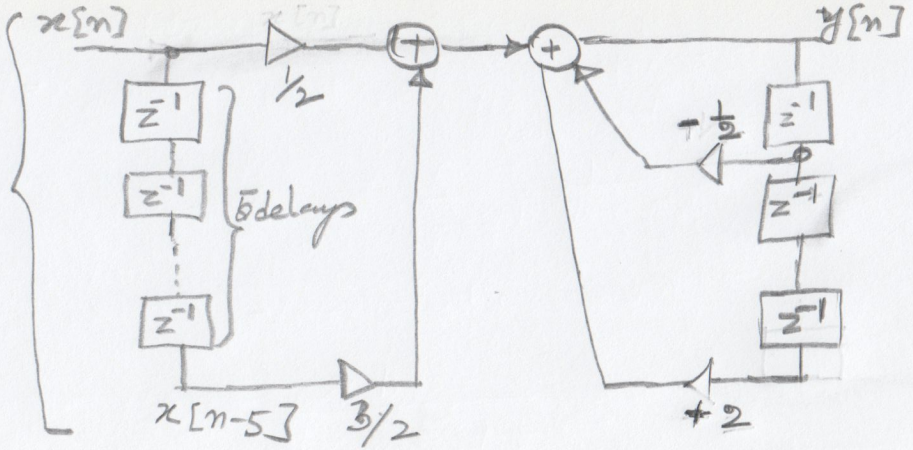
Topic 2: Tutorial Questions (Level 2)

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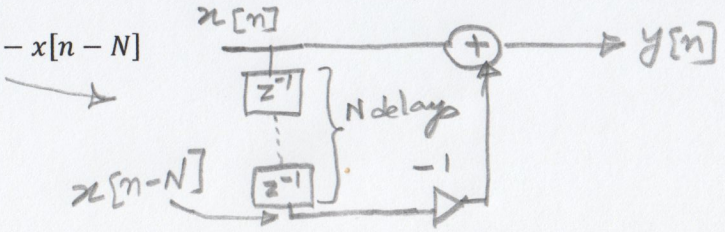
3. Draw a system implementation for each of the following equations:

a) $2y[n] + y[n-1] - 4y[n-3] = x[n] + 3x[n-5]$

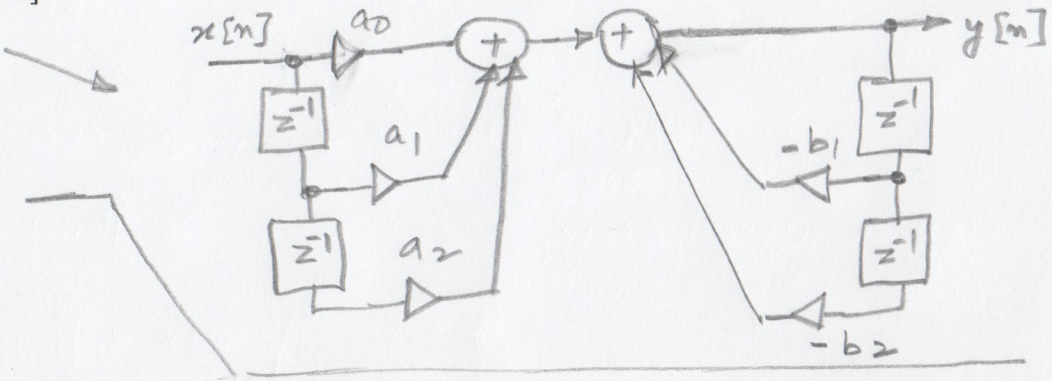
$y[n] = \frac{1}{2}x[n] + \frac{3}{2}x[n-5] - \frac{1}{2}y[n-1] + 2y[n-3]$



b) $y[n] = x[n] - x[n-N]$



c) $y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] - b_1y[n-1] - b_2y[n-2]$



4. A difference equation for a particular filter is given by

$y[n] = 0.1x[n] - 0.1x[n-2] + 0.8x[n-3] + 0.1x[n-4] + 0.6x[n-6]$

Find the impulse response of the above filter.

This is a FIR filter.

$\therefore h[n] = \{0.1, 0, -0.1, 0.8, 0.1, 0, 0.6\}$ & $h[n] = 0,\ n \geq 7$

$n = 0$

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5. Consider the system described by the difference equation

$$y[n] = ay[n-1] + bx[n]$$

Determine 'b' in terms of 'a' so that $\sum_{n=-\infty}^{\infty} h[n] = 1$.

Let $x[n] = \delta[n]$ then $y[n] = h[n]$

$\delta[0] = 1$

$n=0 \quad y[0] = a\,y[-1] + b\,x[0] = b$

$n=1 \quad y[1] = a\,y[0] + b\,x[1] = ab$

$n=2 \quad y[2] = a\,y[1] + b\,x[2] = a^2 b$

$n = n-1 \quad y[n-1] = a^{n-1} b$

$n = n \quad \boxed{y[n] = a^n b \quad n \geq 0}$

$$\sum_{n=-\infty}^{\infty} h[n] = \sum_{n=0}^{\infty} a^n b = b\left[1 + a + a^2 + \cdots\right] = b \cdot \frac{1}{1-a}$$

$$\sum_{n=-\infty}^{\infty} h[n] = 1 \text{ (given)} \quad \therefore b \cdot \frac{1}{1-a} = 1 \Rightarrow \boxed{b = 1-a}$$

6. Let $x[n] = \{1\ 4\ 0\ 2\}$ and $h[n] = \{1\ 2\ 1\}$. Find their convolution (Both sequences start at n=0).

$$X(z) = 1 + 4z^{-1} + 2z^{-3}; \quad H(z) = 1 + 2z^{-1} + z^{-2}$$

$$Y(z) = X(z) \cdot H(z) = 1 + 2z^{-1} + z^{-2} + 4z^{-1} + 8z^{-2} + 4z^{-3} + 2z^{-3} + 4z^{-4} + 2z^{-5}$$

$$Y(z) = 1 + 6z^{-1} + 9z^{-2} + 6z^{-3} + 4z^{-4} + 2z^{-5}$$

Take inverse z-transform $\Rightarrow y[n] = x(n) * h[n] = \{1, 6, 9, 6, 4, 2\}$

7. Find the impulse response h[n] for the following system:

$$H(z) = \frac{1 - \frac{1}{2}z^{-1}}{\left[1 - \frac{1}{4}z^{-1}\right]\left[1 + \frac{1}{4}z^{-1}\right]}$$

$$\frac{1 - \frac{1}{2}z^{-1}}{\left[1 - \frac{1}{4}z^{-1}\right]\left[1 + \frac{1}{4}z^{-1}\right]} = \frac{A}{\left[1 - \frac{1}{4}z^{-1}\right]} + \frac{B}{1 + \frac{1}{4}z^{-1}}$$

To find A, substitute $z^{-1} = 4$ $\{$i.e. $1 - \frac{1}{4}z^{-1} = 0\}$ $\rightarrow$

$$A = \left.\frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{1}{4}z^{-1}}\right|_{z^{-1}=4} = \frac{1 - \frac{1}{2} \times 4}{1 + \frac{1}{4} \times 4} = -\frac{1}{2}$$

Similarly $B = \left.\frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{4}z^{-1}}\right|_{z^{-1}=-4} = \frac{1 - \frac{1}{2}(-4)}{1 - \frac{1}{4}(-4)} = \frac{3}{2}$

$$\therefore H(z) = \frac{-\frac{1}{2}}{1 - \frac{1}{4}z^{-1}} + \frac{\frac{3}{2}}{1 + \frac{1}{4}z^{-1}}$$

$$\therefore h[n] = -\frac{1}{2}\left(\frac{1}{4}\right)^n u[n] + \frac{3}{2}\left(-\frac{1}{4}\right)^n u[n]$$

Topic 2: Tutorial Questions (Level 2)

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8. Consider the following system:

$$H(z) = \frac{1 - 2z^{-1} + 2z^{-2} - z^{-3}}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

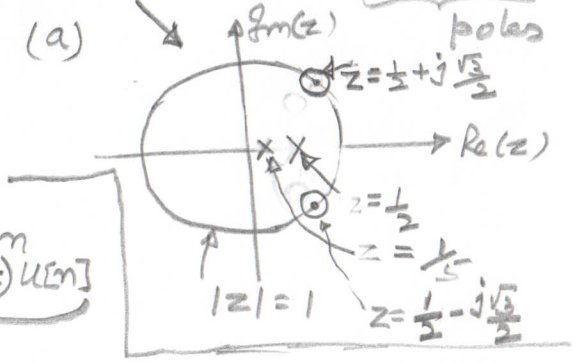
a) Sketch the pole-zero pattern. Is this system stable?

b) Determine the impulse response of the system.

$= \frac{(1 - z^{-1} + z^{-2})(1 - z^{-1})}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})} = \frac{1 - z^{-1} + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{5}z^{-1})} = \frac{z^2 - z + 1}{(z - \frac{1}{2})(z - \frac{1}{5})}$

zeros: $z = \frac{1 \pm j\sqrt{3}}{2}$

(a)



(b) $H(z) = 1 + \left[\frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - \frac{1}{5}z^{-1}}\right] z^{-1}$ ← similar to Q7, we can find $A = \frac{5}{2}$, $B = \frac{-14}{5}$

$H[z] = 1 + \frac{(5/2)z^{-1}}{1 - \frac{1}{2}z^{-1}} - \frac{(14/5)z^{-1}}{1 - \frac{1}{5}z^{-1}}$

Inverse z-transform

$\therefore h[n] = \delta[n] + \frac{5}{2}\left(\frac{1}{2}\right)^{n-1}u[n-1] - \left(\frac{14}{5}\right)\left(\frac{1}{5}\right)^{n-1}u[n-1] = \underbrace{\delta[n] + 5\left(\frac{1}{2}\right)^{n}u[n] - 14\left(\frac{1}{5}\right)^{n}u[n]}_{= h[n]}$

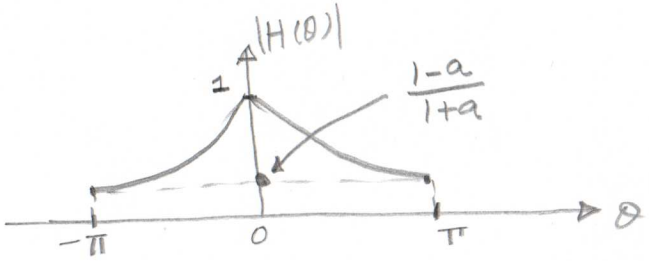
9. Determine and sketch the approximate magnitude response for the following filter: $H(z) = \frac{1-a}{1-az^{-1}}$ $0<a<1$

$H(\theta) = H(z)\Big|_{z = e^{j\theta}} = \frac{1-a}{1-ae^{-j\theta}}$

$|H(\theta)|^2 = H(\theta) \cdot H^*(\theta) = \frac{1-a}{1-ae^{-j\theta}} \cdot \frac{1-a}{1-ae^{j\theta}} = \frac{(1-a)^2}{1 - a(e^{j\theta} + e^{-j\theta}) + a^2} = \frac{(1-a)^2}{1 - 2a\cos\theta + a^2}$

Complex conjugate

$\therefore |H(\theta)| = \frac{1-a}{\sqrt{1 - 2a\cos\theta + a^2}}$



Topic 2: Tutorial Questions (Level 2)

SOLO – Level 2

10. A speech signal is sampled with a sampling period of 125ms. A frame of 256 samples is selected and a 256 point DFT is computed. What is the spacing between the DFT values in Hz.

$f_s = \frac{1}{125\times10^{-3}}$ $\quad N = 256$

$\Delta f = \frac{f_s}{N} = \frac{1}{125\times10^{-3}} \cdot \frac{1}{256} = 0.03125\,Hz$

10. Let x[n]={1, 2, 1, 0}. With N=4, compute the 4-point DFT.

$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N} n.k}$; $N=4$, $\therefore X[k] = \sum_{n=0}^{3} x[n]\, e^{-j\frac{2\pi}{4} nk}$

$X[k] = x[0] + x[1]e^{-j\frac{\pi}{2}k} + x[2]e^{-j\pi k} + x[3]e^{-j\frac{3\pi}{2}k}$

$\quad\quad = 1 \quad\quad 2 \quad\quad\quad 1 \quad\quad\quad 0$

$\boxed{X[k] = 1 + 2e^{-j\frac{\pi}{2}k} + e^{-j\pi k}}$

$k=0$ $X[0] = 4$; $k=1$ $X[1] = 1 + 2e^{-j\frac{\pi}{2}} + e^{-j\pi} = -j2$

$k=2$ $X[2] = 1 + 2e^{-j\pi} + e^{-j2\pi} = 0$

$k=3$ $X[3] = 1 + 2e^{-j\frac{3\pi}{2}} + e^{-j3\pi} = j2$

$\boxed{X[k] = [4, -j2, 0, j2]}$

12. Find the 8-point DFT of $x[n]$ ={1, 1, 0, 0, 0, 0, 0, 0}, using conjugate symmetry property of $X[k]$.

Since x[2] to x[7] are all zeros, the upper index in the DFT summation will be n=1

$X[k] = \sum_{n=0}^{1} x[n]\, e^{-j\frac{2\pi}{8} nk}$ {N=8} ; $X[k] = x[0] + x[1]e^{-j\frac{2\pi}{8}k} = 1 + e^{-j\frac{\pi}{4}k}$ for $k = 0, 1, 2, \cdots 7$

N=8;

$X[k] = 1 + e^{-j\frac{\pi}{4}k}$

$k=0$ $X[0] = 1+1 = 2$

$k=1$ $X[1] = 1 + e^{-j\frac{\pi}{4}} = 1.707 - j0.707$

$k=2$ $X[2] = 1 + e^{-j\frac{\pi}{2}} = 1 - j$

$k=3$ $X[3] = 1 + e^{-j\frac{3\pi}{4}} = 0.293 - j0.707$

$k=4$ $X[0] = 1 + e^{-j\pi} = 0$

Using conjugate symmetry property

$X[5] = X[3] = 0.293 - j0.707$

$X[6] = X[2] = 1 - j$

$X[7] = X[1] = 1.707 - j0.707$

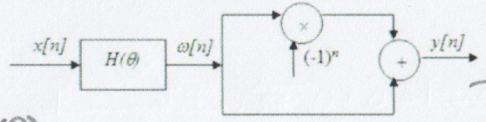
$X[0]$ $X[1]$ $X[2]$ $X[3]$ $X[4]$ $X[5]$ $X[6]$ $X[7]$

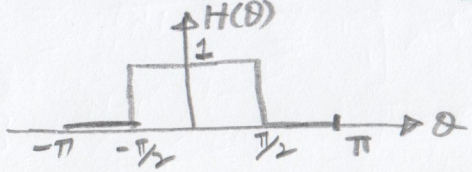
Topic 2: Tutorial questions (Levels 2)

SOLO – Level 2

13. For the system in the figure below, sketch the output *y[n]* when the input *x[n]* is *δ[n]* and *H(θ)* is an ideal lowpass filter as follows:

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \le \pi \end{cases}$$





$w[n] = h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)\, e^{j\theta n}\, d\theta = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} 1\, e^{j\theta n}\, d\theta$

$h[n] = \frac{1}{2\pi}\,\frac{e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}}{jn} = \frac{1}{\pi}\,\frac{\sin n\frac{\pi}{2}}{n}$

$y[n] = (-1)^n h[n] + h[n] = h[n]\left\{1 + (-1)^n\right\}$

$y[n] = \frac{1}{2}\,\frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}\left\{1 + (-1)^n\right\}$

$$y[n] = \frac{1}{2}\left\{1 + (-1)^n\right\} \cdot \frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}$$

w[n] becomes h[n] when x[n] = δ[n]

14. A 3- second signal is sampled at f_s = 100 Hz. The maximum spectral spacing is to be $\Delta f = 0.25\ Hz$. How many samples are needed for the DFT and FFT ?

Number of samples $N = \frac{f_s}{\Delta f} = \frac{100}{0.25} = 400$

3s signal gives 300 samples. To make 400 samples, we must pad the signal with 100 zeros.

DFT : 400 samples

FFT : 512 samples (2^9) ie another 112 zeros must be added. ie spacing $\Delta f = \frac{100}{512} = 0.1953\ Hz$ — better than required, resolution of 0.25 Hz

15. Classify the following system in terms of its linearity, time invariance, memory and causality: $y[n] = |x[-n]|$

answer : non-linear, time invariant, dynamic, non-causal

Linearity : $x[n] \xrightarrow{H} y[n] = |x[-n]|$

$x_1[n] \xrightarrow{H} y_1[n] = |x_1[-n]|$

$x_2[n] \xrightarrow{H} y_2[n] = |x_2[-n]|$

$a_1 y_1[n] + a_2 y_2[n] = a_1|x_1[-n]| + a_2|x_2[-n]|$ — ①

$a_1 x_1[n] + a_2 x_2[n] \xrightarrow{H} |a_1 x_1[-n] + a_2 x_2[-n]|$ — ②

① & ② are not same

Non-linear

Time invariant

$x(n-n_0) \xrightarrow{H} |x[-(n-n_0)]|$ — ③

$y(n-n_0) \xrightarrow{H} |x[-(n-n_0)]|$ — ④

③ & ④ are same

So time invariant

Memory

If $n=2$, $y[2] = |x[-2]|$

output relies on the past input

∴ dynamic

causal

If $n=-2$, $y[-2] = |x[2]|$

output relies on future input

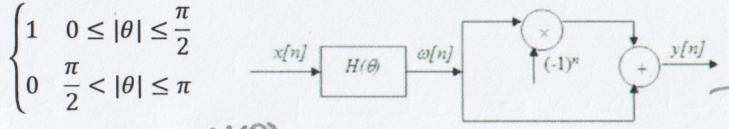
∴ non-causal

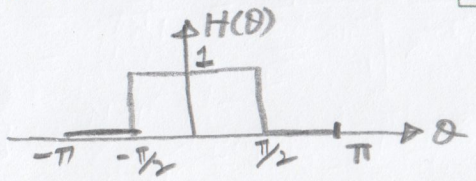
Topic 2: Tutorial questions (Levels 2)

SOLO – Level 2

13. For the system in the figure below, sketch the output *y[n]* when the input *x[n]* is $\delta[n]$ and $H(\theta)$ is an ideal lowpass filter as follows:

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \le \pi \end{cases}$$





$w[n] = h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)\, e^{j\theta n}\, d\theta = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} 1\, e^{j\theta n}\, d\theta$

$h[n] = \frac{1}{2\pi}\,\frac{e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}}{jn} = \frac{1}{\pi}\,\frac{\sin n\frac{\pi}{2}}{n}$

$y[n] = (-1)^n h[n] + h[n] = h[n]\left\{1 + (-1)^n\right\}$

$y[n] = \frac{1}{2}\,\frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}\left\{1 + (-1)^n\right\}$

$$y[n] = \frac{1}{2}\left\{1 + (-1)^n\right\} \cdot \frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}$$

w[n] becomes h[n] when x[n] = δ[n]

14. A 3- second signal is sampled at f_s = 100 Hz. The maximum spectral spacing is to be $\Delta f = 0.25\ Hz$. How many samples are needed for the DFT and FFT ?

Number of samples $N = \frac{f_s}{\Delta f} = \frac{100}{0.25} = 400$

3s signal gives 300 samples. To make 400 samples, we must pad the signal with 100 zeros.

DFT : 400 samples

FFT : 512 samples (2^9) i.e another 112 zeros must be added. i.e spacing $\Delta f = \frac{100}{512} = 0.1953\ Hz$

better than required, resolution of 0.25 Hz

15. Classify the following system in terms of its linearity, time invariance, memory and causality: $y[n] = |x[-n]|$

Answer : non-linear, time invariant, dynamic, non-causal

Linearity : $x[n] \xrightarrow{H} y[n] = |x[-n]|$

$x_1[n] \xrightarrow{H} y_1[n] = |x_1[-n]|$

$x_2[n] \xrightarrow{H} y_2[n] = |x_2[-n]|$

$a_1 y_1[n] + a_2 y_2[n] = a_1\,|x_1[-n]| + a_2\,|x_2[-n]|$ — ①

$a_1 x_1[n] + a_2 x_2[n] \xrightarrow{H} |a_1 x_1[-n] + a_2 x_2[-n]|$ — ②

① & ② are not same

Non-linear

Time invariant

$x(n-n_0) \xrightarrow{H} |x[-(n-n_0)]|$ — ③

$y(n-n_0) \xrightarrow{H} |x[-(n-n_0)]|$ — ④

③ & ④ are same

So time invariant

Memory

If $n=2$, $y[2] = |x[-2]|$

output relies on the past input

∴ Dynamic

Causal

If $n=-2$, $y[-2] = |x[2]|$

output relies on future input

∴ non-causal

Topic 2: Tutorial questions (Level 3)

SOLO – Level 3

1. If $y[n] = x[n] * h[n]$, then show that $Y(z) = X(z) \cdot H(z)$.

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k] \quad ; \quad Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n}$$

← Definition of z-transform

$$\therefore Y(z) = \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} h[k]\, x[n-k] \right] z^{-n} = \sum_{n=-\infty}^{\infty} \left[h[k] \cdot z^{-k} \cdot \sum_{k=-\infty}^{\infty} x[n-k]\, z^{k} \cdot z^{-n} \right]$$

$$Y(z) = \sum_{k=-\infty}^{\infty} \left\{ h[k] z^{-k} \cdot \sum_{n=-\infty}^{\infty} x[n-k]\, z^{-(n-k)} \right\}$$

{Multiplying by $z^{k} \cdot z^{-k}$ and changing the order of the summation]

By substituting $m = n - k$

$$Y(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-k} \cdot \sum_{m=-\infty}^{\infty} x[m]\, z^{-m}$$

$= H(z)$ $\quad$ $= X(z)$

$$\Rightarrow Y(z) = H(z) \cdot X(z)$$

$$Y(z) = X(z) \cdot H(z)$$

2. Find the z-transform, G(z), if $g[n] = \begin{cases} na^n & n \geq 0 \\ 0 & n < 0 \end{cases}$ where a is a constant

Let $x[n] = a^n \Rightarrow X(z) = \frac{z}{z-a} \quad \therefore \frac{dX(z)}{dz} = \frac{-a}{(z-a)^2}$

$g[n] = n a^n$

$G(z) = -z \frac{dX(z)}{dz}$ ←

$$G(z) = -z \cdot \frac{-a}{(z-a)^2} = \frac{a z}{z^2 (1 - a z^{-1})^2} = \frac{a z^{-1}}{(1 - a z^{-1})^2}$$

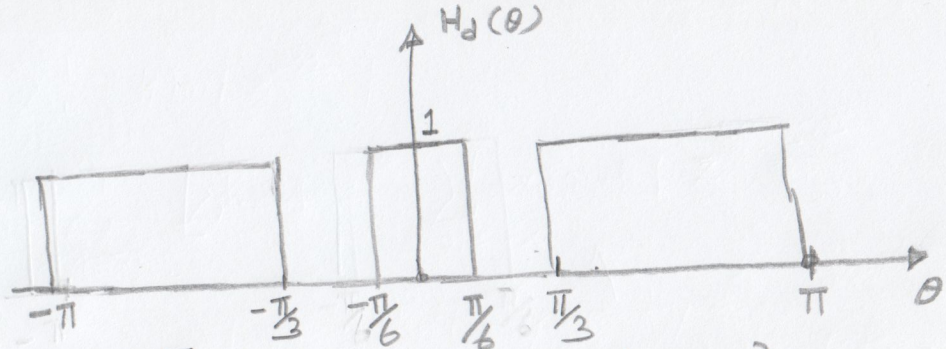
$$G(z) = \frac{a z^{-1}}{(1 - a z^{-1})^2}$$

Topic 2: Tutorial questions (Level 3)

SOLO – Level 3

3. The frequency response of a band stop filter is given by

$$H_d(\theta) = \begin{cases} 1 & |\theta| \le \frac{\pi}{6} \\ 0 & \frac{\pi}{6} < |\theta| < \frac{\pi}{3} \\ 1 & \frac{\pi}{3} \le |\theta| \le \pi \end{cases}$$



Show that the impulse response $h_d[n]$ of the band stop filter is given by

$$h_d[n] = \delta[n] - \frac{\sin\left(\frac{\pi}{3}n\right)}{\pi n} + \frac{\sin\left(\frac{\pi}{6}n\right)}{\pi n}$$

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\theta)e^{jn\theta} = \frac{1}{2\pi}\left\{\int_{-\pi}^{-\pi/3} 1\cdot e^{jn\theta}d\theta + \int_{-\pi/6}^{\pi/6} 1\cdot e^{jn\theta}d\theta + \int_{\pi/3}^{\pi} 1\cdot e^{jn\theta}d\theta\right\}$$

$$= \frac{1}{2\pi}\left\{\left[\frac{e^{jn\theta}}{jn}\right]_{-\pi}^{-\pi/3} + \left[\frac{e^{jn\theta}}{jn}\right]_{-\pi/6}^{\pi/6} + \left[\frac{e^{jn\theta}}{jn}\right]_{\pi/3}^{\pi}\right\}$$

$$= \frac{1}{2\pi}\left\{\frac{e^{-j\frac{\pi}{3}n} - e^{-j\pi n}}{jn} + \frac{e^{j\frac{\pi}{6}n} - e^{-j\frac{\pi}{6}n}}{jn} + \frac{e^{j\pi n} - e^{j\frac{\pi}{3}n}}{jn}\right\}$$

$$= \frac{1}{\pi n}\left\{-\underbrace{\frac{(e^{j\frac{\pi}{3}n} - e^{-j\frac{\pi}{3}n})}{2j}}_{-\sin\frac{\pi}{3}n} + \underbrace{\frac{e^{j\frac{\pi}{6}n} - e^{-j\frac{\pi}{6}n}}{2j}}_{=\sin\frac{\pi}{6}n} + \underbrace{\frac{e^{j\pi n} - e^{-j\pi n}}{2j}}_{=\sin\pi n}\right\}$$

$$= \frac{\sin\pi n}{\pi n} - \frac{\sin\frac{\pi}{3}n}{\pi n} + \frac{\sin\frac{\pi}{6}n}{\pi n}$$

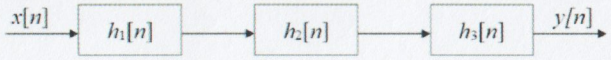
note:
$\frac{\sin n\pi}{\pi n} = 1$ when $n=0$
& zero elsewhere
$\therefore \delta[n]$

$$h_d[n] = \delta[n] - \frac{\sin\frac{\pi}{3}n}{\pi n} + \frac{\sin\frac{\pi}{6}n}{\pi n}$$

Topic 2: Tutorial questions (Level 3)

SOLO – Level 3

4. Three systems with the following impulse responses are connected in cascade: $h_1[n] = \delta(n) - \delta(n-1)$; $h_2[n] = g[n]$; $h_3[n] = u[n]$; where $u[n]$ is the unit step sequence and $\delta[n]$ is the unit sample sequence (i.e an impulse).



What is the impulse response $h_c(n)$ of the overall system?

$h_1[n] = \delta[n] - \delta[n-1] \Rightarrow H_1(z) = 1 - z^{-1}$

$h_3[n] = u[n] \Rightarrow H_3(z) = \frac{1}{1-z^{-1}}$

$y[n] = x[n] * h_1[n] * h_2[n] * h_3[n]$

$\frac{Y(z)}{X(z)} = H_1(z) \cdot H_2(z) \cdot H_3(z)$

$H_c(z) = (1 - z^{-1}) \cdot G(z) \cdot \frac{1}{1-z^{-1}}$

$H_c(z) = G(z)$

$\therefore h_c[n] = g[n]$

5. Show that the convolution of the two infinite duration sequences

$$p[n] = a^n u[n] \quad \text{and} \quad q[n] = b^n u[n]$$

for all n, where $u[n]$ is the unit step function and $a \neq b$, is given by

$$y[n] = \frac{a^{n+1} - b^{n+1}}{a - b}$$

$y[n] = p[n] * q[n] = \sum_{k=-\infty}^{\infty} a^k u[k] \cdot b^{n-k} u[n-k]$

$u[k] = 1$ when $k \geq 0$; $u[n-k] = 1$ when $n - k \geq 0$, i.e. $k \leq n$

$= \sum_{k=0}^{n} a^k b^{n-k}$

$= \sum_{k=0}^{n} \left(\frac{a}{b}\right)^k \cdot b^n$

$= b^n \sum_{k=0}^{n} \left(\frac{a}{b}\right)^k$

$y[n] = b^n \sum_{k=0}^{n} \left(\frac{a}{b}\right)^k$

$= b^n \left[1 + \left(\frac{a}{b}\right) + \left(\frac{a}{b}\right)^2 + \cdots + \left(\frac{a}{b}\right)^n\right]$

$= b^n \frac{1 - \left(\frac{a}{b}\right)^{n+1}}{1 - \frac{a}{b}}$

$= \frac{b^n}{b^{n+1}} \frac{b^{n+1} - a^{n+1}}{b - a} = \frac{b^n \times b}{b^{n+1}} \cdot \frac{a^{n+1} - b^{n+1}}{a - b}$

$y[n] = \frac{a^{n+1} - b^{n+1}}{a - b}$

Topic 2: Tutorial questions (Level 3)

SOLO – Level 3

6. Consider a system with an input x[n] and an output y[n] which satisfies the following difference equation:

$$y[n] = n\,y[n-1] + x[n]$$

If $x[n] = \delta[n]$, determine $y[n]$ for all values of n.

$x[n] = \delta[n]$ $\quad = 1$ for $n = 0$

$n = 0 \quad y[0] = 0\,y[-1] + x[0] = 1$

$n = 1 \quad y[1] = 1\,y[0] + x[1]_{=0} = 1 \cdot 1$

$n = 2 \quad y[2] = 2[1] + x[2]_{=0} = 2 \cdot 1$

$n = 3 \quad y[3] = 3[2 \times 1] + x[3]_{=0} = 3 \cdot 2 \cdot 1$

$n = 4 \quad y[4] = 4[3 \times 2 \times 1] + x[4]_{=0} = 4 \cdot 3 \cdot 2 \cdot 1$

$n = n \quad y[n] = n \cdot (n-1)(n-2)(n-3) \cdots 1 = n!$

$$\boxed{y[n] = n! \quad n \geq 1}$$

Topic 2: Tutorial questions (Level 4)

SOLO – Level 4

1. A system L is known to be linear. The responses of L to inputs $x_1[n]$ and $x_2[n]$ are given by $y_1[n]$ and $y_2[n]$ respectively (see Figure 1).

   (a) Determine whether the system L could be time-invariant.

   (b) If the input x[n] to the system is $\delta[n]$, what is the system response y[n]?

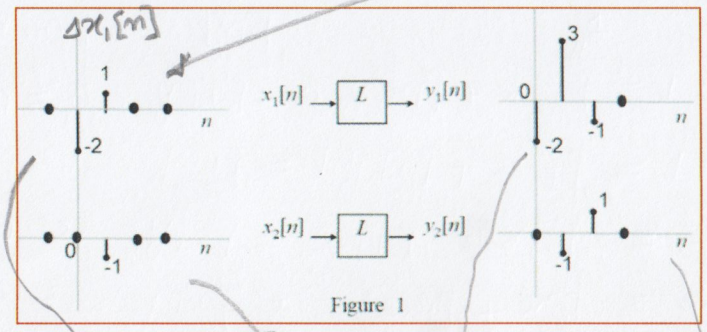


Figure 1

(a) $x_1[n]$ can be formulated from the graph as follows:

$x_1[n] = -x_2[n] + 2x_2[n+1]$

If the system is time invariant, then

$y_1[n] = -y_2[n] + 2y_2[n+1]$

$[0, 1, -1]$ $[-2, 2, 0]$

$y_1[n] = \{(0-2), (1+2), (-1+0)\}$

$= \{-2, 3, -1\}$

Same as the output of $y_1[n]$ shown in the figure

$\therefore$ L is time-invariant

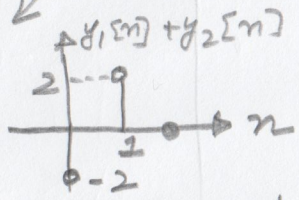
(b) add $x_1[n] + x_2[n] = -2\delta[n]$

$-2\delta[n]$ → L → $y_1[n] + y_2[n] = -2h[n]$

$h[n]$

$-2\delta[n] \xrightarrow{L} -2h[n]$

$\delta[n] \longrightarrow h[n]$



Divide it by -2, then you get $h[n]$

$\therefore h[n] = \{1, -1\}$

Topic 2: Tutorial questions (Level 4)

SOLO – Level 4

2. Consider the interconnection of LTI systems as shown in Figure 2:

   (a) Express the overall impulse response in terms of $h_1[n]$, $h_2[n]$, $h_3[n]$ and $h_4[n]$.

   (b) Determine *h[n]* when

   $h_1[n] = \{\frac{1}{2}\ \underset{\uparrow}{\frac{1}{4}}\ \frac{1}{2}\}$; $h_2[n] = h_3[n] = (n+1)\,u[n]$; $h_4[n] = \delta[n-2]$

   (c) Determine the response of the system in part(b) if

   $x[n] = \delta[n+2] + 3\delta[n-1] - 4\delta[n-3]$

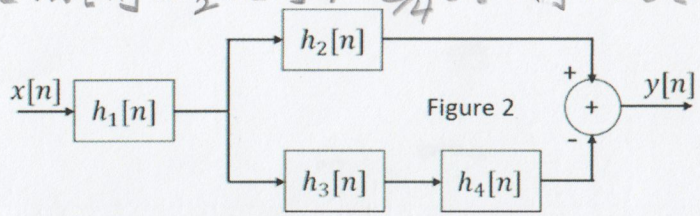


Figure 2

(a) $h[n] = h_1[n] * [h_2[n] - h_3[n] * h_4[n]]$

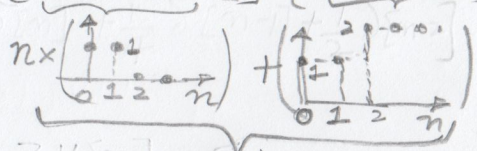
(b) $h_3[n] * h_4[n] = (n+1)\,u[n] * \delta[n-2]$

$= (n-1)\,u[n-2]$

$h_2[n] - h_3[n] * h_4[n]$

$= (n+1)\,u[n] - (n-1)\,u[n-2]$

$= n\{u[n] - u[n-2]\} + \{u[n] + u[n-2]\}$

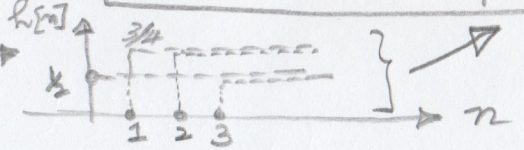


$= 2u[n] - \delta[n]$

Given (see (b)) $h_1[n] = \frac{1}{2}\delta[n] + \frac{1}{4}\delta[n-1] + \frac{1}{2}\delta[n-2]$

Hence $h[n] = \{\frac{1}{2}\delta[n] + \frac{1}{4}\delta[n-1] + \frac{1}{2}\delta[n-2]\} * \{2u[n] - \delta[n]\}$

$h[n] = \frac{1}{2}\delta[n] + \frac{5}{4}\delta[n-1] + 2\delta[n-2] + \frac{5}{2}u[n-3]$



$x[n] = \{1, 0, 0, \underset{\uparrow}{3}, 0, -4\}$ ($n=0$)

$h[n] = \frac{1}{2}\delta[n] + \frac{5}{4}\delta[n-1] + 2\delta[n-2] + \frac{5}{2}u[n-3]$

$y[n] = x[n] * h[n]$

$= \{\frac{1}{2}, \frac{5}{4}, \underset{\uparrow}{2}, \frac{25}{4}, \frac{13}{2}, 5, 2, 0, 0 \ldots\}$ ($n=0$)

Note

$H(z) = \left[\frac{1}{2} + \frac{1}{4}z^{-1} + \frac{1}{2}z^{-2}\right] \cdot \left[\frac{2}{1-z^{-1}}\right]$

$= \frac{\frac{1}{2} + \frac{3}{4}z^{-1} + \frac{3}{4}z^{-2} + \frac{1}{2}z^{-3}}{1-z^{-1}}$

$h[n] = \frac{1}{2}u[n] + \frac{3}{4}u[n-1] + \frac{3}{4}u[n-2] + \frac{1}{2}u[n-3]$

Topic 2: Tutorial questions (Level 4)

SOLO – Level 4

3. Find a difference equation to implement a system that has a unit impulse response:

$$h[n] = (\tfrac{1}{4})^n \cos(n\tfrac{\pi}{3})\, u[n]$$

Note: To find the difference equation, we must first find the z-transform response $H(\theta)$

Using the z-transform table

$$H(z) = \frac{1 - z^{-1}(\frac{1}{4})\cos(\frac{\pi}{3})}{1 - z^{-1}2(\frac{1}{4})\cos\frac{\pi}{3} + (\frac{1}{4})^2 z^{-2}} = \frac{1 - \frac{1}{8}z^{-1}}{1 - \frac{1}{4}z^{-1} + \frac{1}{16}z^{-2}} = \frac{Y(z)}{X(z)}$$

$$Y(z)\left[1 - \frac{1}{4}z^{-1} + \frac{1}{16}z^{-2}\right] = X(z)\left[1 - \frac{1}{8}z^{-1}\right]$$

$$\therefore y[n] = x[n] - \frac{1}{8}x[n-1] + \frac{1}{4}y[n-1] - \frac{1}{16}y[n-2]$$

4. An L[th] order moving average filter is a linear time invariant system that, for an input $x[n]$, produces the output $y[n] = \frac{1}{L+1}\sum_{k=0}^{L} x[n-k]$. Show that the frequency response of this system is $H(\theta) = \frac{1}{L+1} e^{-jL\theta/2} \frac{\sin[(L+1)\theta/2]}{\sin(\frac{\theta}{2})}$

If the input $x[n] = \delta[n]$, the response, by definition will be the impulse response $h[n]$

$$\therefore h[n] = \frac{1}{L+1}\sum_{k=0}^{L}\delta[n-k] \Rightarrow H(\theta) = \frac{1}{L+1}\sum_{k=0}^{L} e^{-jk\theta}$$

$$H(\theta) = \frac{1}{L+1}\left[1 + e^{-j\theta} + e^{-j2\theta} + e^{-j3\theta} + \cdots + e^{-jL\theta}\right] = \frac{1}{L+1}\cdot\frac{1 - e^{-j(L+1)\theta}}{1 - e^{-j\theta}}$$

(using Geometric series)

Factoring out $e^{-j\frac{(L+1)\theta}{2}}$ from the numerator and a term $e^{-j\frac{\theta}{2}}$ from the denominator, we have

$$H(\theta) = \frac{1}{L+1} e^{-jL\theta/2} \cdot \frac{e^{j\frac{(L+1)\theta}{2}} - e^{-j\frac{(L+1)\theta}{2}}}{e^{+j\theta/2} - e^{-j\theta/2}} = \frac{1}{L+1} \cdot e^{-\frac{jL\theta}{2}} \cdot \frac{\sin(L+1)\theta/2}{\sin\theta/2}$$

Topic 2: Tutorial questions (Level 5)

SOLO – Level 5

1. Consider a bandpass filter with a spectrum X(θ) as shown in Figure A. Show that x[n] can be formulated as:

$$x[n] = (k_p + k_q)\cos(n\theta_0) - k_p\cos(n\theta_u) - k_q\cos(n\theta_l)$$

$$k_p = \frac{1}{\pi n^2(\theta_u - \theta_0)}; \quad k_q = \frac{1}{\pi n^2(\theta_0 - \theta_l)}; \quad \theta = 2\pi\frac{f}{f_s}$$

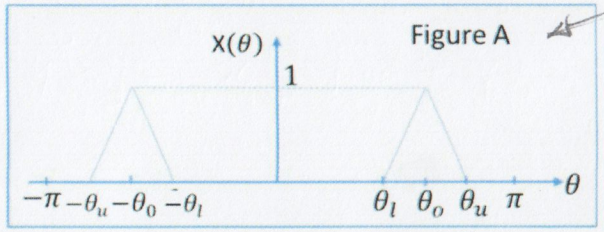
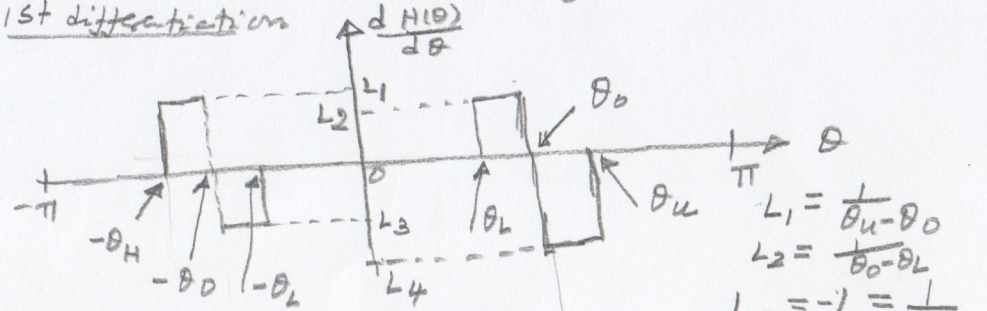


Figure A

note: $\theta_u \neq \theta_0$, $\theta_0 \neq \theta_L$

To find impulse response, we take IDTFT of $X(\theta)$. Use differentiation property of DTFT. $n\,h[n] \Leftrightarrow j\frac{dH(\theta)}{d\theta}$

1st differentiation $\frac{d\,H(\theta)}{d\theta}$



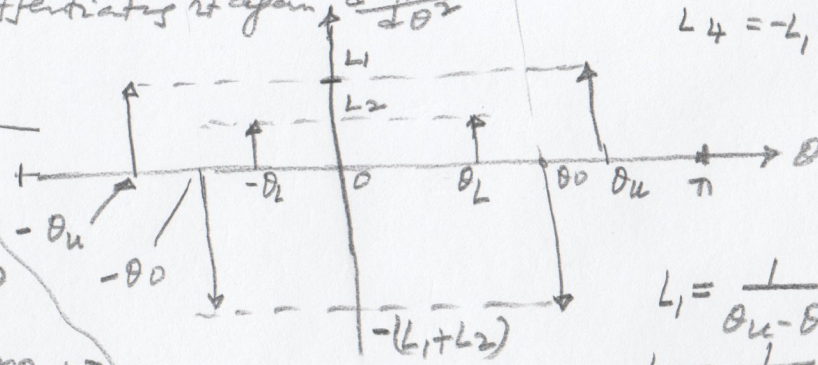
$L_1 = \frac{1}{\theta_u - \theta_0}$

$L_2 = \frac{1}{\theta_0 - \theta_L}$

$L_3 = -L_2 = \frac{1}{\theta_L - \theta_0}$

$L_4 = -L_1 = \frac{1}{\theta_0 - \theta_u}$

Differentiating it again $\frac{d^2H(\theta)}{d\theta^2}$



$L_1 = \frac{1}{\theta_u - \theta_0}$

$L_2 = \frac{1}{\theta_0 - \theta_L}$

Taking IDFT of $\frac{d^2H(\theta)}{d\theta^2} = \frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{d^2H(\theta)}{d\theta^2}e^{jn\theta}\,d\theta$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi}\Big[L_2\{\delta(\theta-\theta_L)+\delta(\theta+\theta_L)\} + (L_1+L_2)\{\delta(\theta-\theta_0)+\delta(\theta+\theta_0)\} + L_1\{\delta(\theta-\theta_u)+\delta(\theta+\theta_u)\}\Big]e^{jn\theta}\,d\theta$$

$$= \frac{1}{2\pi}\Big\{L_2(e^{jn\theta_L}+e^{-jn\theta_L}) - (L_1+L_2)(e^{jn\theta_0}+e^{-jn\theta_0}) + L_1(e^{jn\theta_u}+e^{-jn\theta_u})\Big\}$$

$$= \frac{1}{\pi}\Big\{\underbrace{\frac{1}{\theta_u-\theta_0}}\cos n\theta_u + \frac{1}{\theta_0-\theta_L}\cos n\theta_L - \Big(\frac{1}{\theta_u-\theta_0}+\frac{1}{\theta_0-\theta_L}\Big)\cos n\theta_0\Big\}$$

To get $h[n]$ multiply the above by $\big(\frac{1}{n}\cdot\frac{1}{n} = \frac{1}{n^2}\big)$ ← [note $n\,h[n] \Leftrightarrow j\frac{dH(\theta)}{d\theta}$]

$$\therefore h[n] = \Big(\frac{1}{\theta_u-\theta_0}+\frac{1}{\theta_0-\theta_L}\Big)\cdot\frac{1}{n^2\pi}\cos n\theta_0 - \frac{1}{\theta_u-\theta_0}\cdot\frac{1}{n^2\pi}\cos n\theta_u - \frac{1}{\theta_0-\theta_L}\cdot\frac{1}{n^2\pi}\cos n\theta_L$$

(k_p, k_2, k_p, k_2)

Topic 2: Tutorial questions (Level 5)

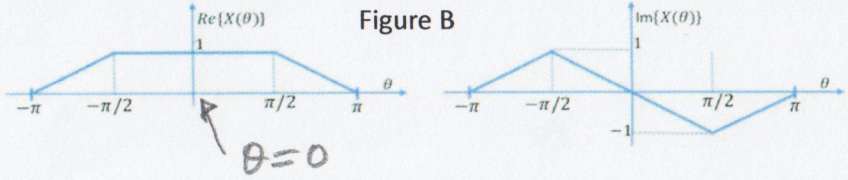
SOLO – Level 5

2. The centre of gravity of a sequence is defined by $c = \frac{\sum_{n=-\infty}^{\infty} n\,x[n]}{\sum_{n=-\infty}^{\infty} x[n]}$ and is used as a measure of the *time delay* of a sequence: Find (using the properties of DTFT)

   (a) an expression for *c* in terms of the DTFT of x[n].

   (b) the value of *c* for the sequence x[n] that has a DTFT as shown in Figure B.

Figure B



To find the value of c in terms of X(θ), first note that the denominator is simply the value of $X(\theta)\big|_{\theta=0}$ (ie DC value)

The Numerator $\sum_{n=-\infty}^{\infty} n\,x[n] = j\frac{d}{d\theta}X(\theta)\Big|_{\theta=0}$ $\quad \left[nx[n] \overset{DTFT}{\Leftrightarrow} j\frac{d}{d\theta}X(\theta)\right]$ see table in the notes

$\therefore c = \dfrac{j\frac{d}{d\theta}X(\theta)\Big|_{\theta=0}}{X(\theta)\Big|_{\theta=0}}$

From Figure B, we have $X(\theta)\big|_{\theta=0} = 1$

$\frac{d}{d\theta}X(\theta) = \frac{d}{d\theta}Re[X(\theta)]\Big|_{\theta=0} + j\frac{d}{d\theta}Im(X(\theta))\Big|_{\theta=0}$

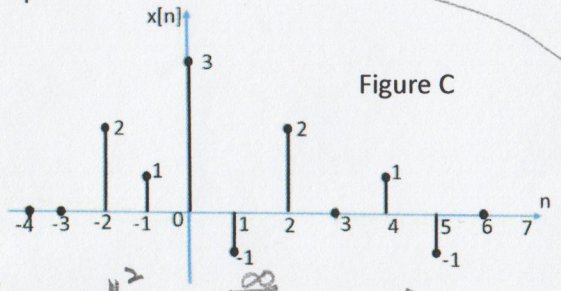
$= 0 + j\left(\frac{-1}{\pi/2}\right)$

(see Figure B gradient) (see Figure B gradient)

$c = j\left(\frac{-2j}{\pi}\right) = \frac{2}{\pi}$

3. For the sequence x[n] plotted in Figure C, evaluate the integral $\int_{-\pi}^{\pi} \left|\frac{d}{d\theta}X(\theta)\right|^2 d\theta$.

Figure C



$\frac{1}{2\pi}\int_{-\pi}^{\pi} |X(\theta)|^2 d\theta = \sum_{n=-\infty}^{\infty} |x(n)|^2$

Topic 2 Page 35

Using Parseval's Theorem

we have $n\,x[n] \overset{DTFT}{\Leftrightarrow} j\frac{d}{d\theta}X(\theta)$ $\therefore \frac{d}{d\theta}X(\theta) \Leftrightarrow -j\,n\,x[n]$

$\therefore \int_{-\pi}^{\pi} \left|\frac{d}{d\theta}X(\theta)\right|^2 = 2\pi \sum_{n=-\infty} |-j\,n\,x[n]|^2 = 2\pi \sum_{n=-\infty}^{\infty} |n\,x[n]|^2 = (75)\times 2\pi$

for x[n] see Figure C